The order of learned saddles in tree tensor networks

and the plateau laws of saddle-to-saddle training

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Abstract

Gradient flow on a tree tensor network (TTN) from small initialization learns a low-rank target one component at a time, resting near a saddle before each new component. How long it rests is governed by the *order* of the saddle: the power at which the loss first decreases away from it. We determine this order exactly at every saddle that represents one learned component. It is the smaller of two graph-theoretic numbers: the size k^* of the smallest Steiner subtree of the tree that connects the cores on which a remaining component differs from the learned one, and twice the smallest level ℓ^* at which a weighted "piggyback" graph built from the remaining components has spectral radius above the learned weight. In particular the order is k^* whenever the remaining components are individually and jointly weaker than the learned one in this spectral sense, which always holds for two components learned in weight order. The order is a property of the tree, so the same target has saddles of different order on different trees. At leading order, the escape from a saddle of order p obeys $\dot{b} = c\, b^{p-1}$, so the plateau lasts $\propto \ln(1/\sigma)$ for $p = 2$ and $\propto \sigma^{-(p-2)}$ for $p \geq 3$; simulations on nine trees confirm these laws.

1 Background and summary

For deep linear networks (DLNs) trained from small initialization, saddle-to-saddle dynamics is well understood: the network learns the singular modes of the target one at a time, and every plateau lasts $\propto \sigma^{-(K-2)}$, where K is the depth and σ the initialization scale (Saxe et al., 2014; Jacot et al., 2022). The exponent has a simple origin. Every layer starts at size σ, the output is a product of all K layers, so a product of K small numbers must grow, and $\dot{a} \propto a^{K-1}$ from $a(0) = \sigma$ blows up after a time $\propto \sigma^{-(K-2)}$. In a DLN every mode needs every layer, so every plateau has the same exponent.

A TTN is also multilinear in K factors (its cores), but each core touches only some of the inputs. After the first component is learned most cores are large, and adding the next component only requires some of them to change from small values. The question is how many, and whether anything else can lower the loss faster. The answer is the order of the saddle at which the flow rests.

The result in words. Let one component j with weight s_j be learned. For each remaining component r, mark the cores carrying the sites where r differs from j, and let k_r be the number of cores in the smallest subtree connecting them (their Steiner subtree). Let $k^* = \min_r k_r$. Then (Theorem 2):

- the loss can always be lowered at order k^*, by opening a new bond channel on the cores of that subtree;

- it can be lowered at a lower order $2\ell < k^*$ exactly when some set of remaining components, each split into two pieces of at most ℓ cores, couples with total strength above s_j (a spectral condition on a weighted graph, Definition 1);
- nothing else lowers the loss faster. The saddle order is $\min(k^*, 2\ell^*)$.

For two components learned in weight order the coupling is too weak, so the order is k^*; for $k^* \le 4$ the order is k^* unless the saddle is strict. When the spectral condition holds, the descent at order $2\ell^*$ uses a direction that is not a teacher component (Example 4).

2 Setting

Tree tensor networks. Let T be a tree with node set V, $|V| = K$. Core $v \in V$ is a tensor G_v with one physical leg of dimension D_v and one bond leg per tree edge at v; bond e has dimension χ_e. Contracting all bond legs gives $W(\theta) \in \bigotimes_v \mathbb{R}^{D_v}$, where θ collects all core entries. We call the input modes *sites*; each core carries the sites of its physical leg ($D_v = d^{n_v}$ for n_v sites of dimension d, and $D_v = 1$ for a core without sites). Examples: the MPS (a chain with one site per core), the binary tree (sites in pairs on the bottom cores), and the hierarchical Tucker (HT) tree (one leaf core per site, internal cores without sites).

Teacher and loss. The network computes $f(x) = \langle W, x_0 \otimes \cdots \otimes x_{N-1} \rangle$, and labels are generated by a teacher tensor W^* in the same way. For whitened Gaussian inputs the population loss is, up to a constant,

$$L(\theta) = \tfrac{1}{2}\|W(\theta) - W^*\|^2. \tag{1}$$

A *pattern teacher* is $W^* = \sum_r s_r U_r$, where each U_r is a tensor product of standard basis vectors, one per site, the patterns are distinct, and $s_r > 0$. A pattern is written as a string of basis indices, e.g. 110010. Distinct patterns are orthonormal.

Learned saddles. Fix a component j. Let $q_v \in \mathbb{R}^{D_v}$ be the pattern of j on the sites of v, so $U_j = \bigotimes_v q_v$. The *learned saddle* θ_S has cores

$$G_v = a_v\, q_v \otimes e_0 \otimes \cdots \otimes e_0, \qquad \textstyle\prod_v a_v = s_j, \tag{2}$$

(e_0 the first basis vector of each bond leg), so $W(\theta_S) = s_j U_j$. Every critical point representing $s_j U_j$ is gauge-equivalent to one of these. Core entries whose bond indices are all 0 are *old*; the others are *new*. The residual is $\mathcal{R} = W^* - s_j U_j = \sum_{r \ne j} s_r U_r$, and $L_S = L(\theta_S) = \frac{1}{2}\|\mathcal{R}\|^2$.

Saddle order. The order of a critical point θ_S is the largest p such that

$$L(\theta) - L_S \;\ge\; -C\,\|\theta - \theta_S\|^p \qquad \text{for all } \theta \text{ near } \theta_S, \tag{3}$$

for some $C > 0$. A strict saddle has order 2. Since L is a polynomial, the order equals the smallest power p for which some curve $\theta(t)$ with $\|\theta(t) - \theta_S\| \le t$ has $L(\theta(t)) - L_S \le -ct^p$ (by the curve selection lemma for semialgebraic sets). Measuring distance to the gauge orbit of θ_S instead gives the same order (the gauge acts linearly and preserves L).

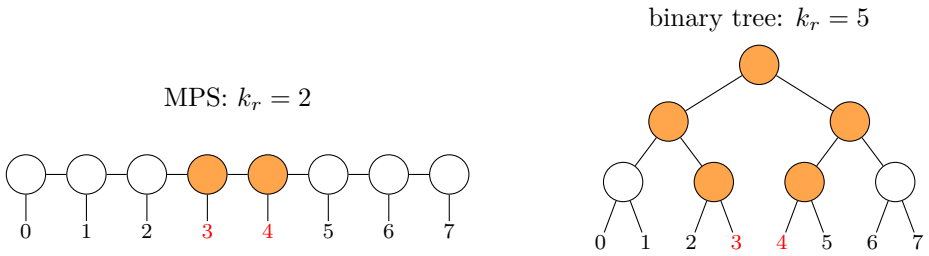


Figure 1: Learned $j = 00000000$, remaining $r = 00011000$: $D_r = \{3, 4\}$ (red sites). Orange: $\mathrm{St}(V_r)$. On an MPS the two sites sit on adjacent cores ($k_r = 2$); on a binary tree the subtree passes through the root ($k_r = 5$); on an HT tree $k_r = 7$.

Steiner subtrees and routes. For a set $B \subseteq V$, $\mathrm{St}(B)$ is the smallest subtree of T containing B, and $\deg B = |\mathrm{St}(B)|$ (so $\deg\{v\} = 1$). For a component $r \neq j$ let D_r be the set of sites where the patterns of r and j differ, and V_r the set of cores carrying them. Define

$$k_r = \deg V_r, \qquad k^* = \min_{r \neq j} k_r. \tag{4}$$

Figure 1 shows an example. θ_S is critical iff $k^* \geq 2$ (Lemma 6).

The piggyback graphs. The second ingredient measures how strongly the remaining components can be fitted jointly by pieces that live on few cores.

Definition 1 (Piggyback graph at level ℓ)**.** For $\ell \geq 1$, the graph $\mathcal{G}_\ell$ has as nodes the pairs (B, π) where $B \subseteq V$ is nonempty with $\deg B \leq \ell$, and π is a pattern on the sites of B that differs from the pattern of j on every core of B. For every component $r \neq j$ and every split $V_r = B_1 \sqcup B_2$ into two nonempty sets with $\deg B_1, \deg B_2 \leq \ell$, it has an edge of weight s_r between $(B_1, \pi_r|_{B_1})$ and $(B_2, \pi_r|_{B_2})$, where $\pi_r|_B$ is the pattern of r on the sites of B. Let Λ_ℓ be the spectral radius of its weighted adjacency matrix $\Gamma^{(\ell)}$ ($\Lambda_\ell = 0$ if it has no edges), and

$$\ell^* = \min\{\ell < k^*/2 : \ \Lambda_\ell > s_j\} \qquad (\ell^* = \infty \text{ if no such } \ell). \tag{5}$$

A node $(\{v\}, \pi)$ stands for rotating the learned vector of core v toward π (old entries); a node with $|B| \geq 2$ stands for new entries on the cores of $\mathrm{St}(B)$ that produce π on B. An edge says that the two pieces together produce component r. $\mathcal{G}_\ell$ is an induced subgraph of $\mathcal{G}_{\ell+1}$, so Λ_ℓ is nondecreasing in ℓ. $\mathcal{G}_1$ is the classical piggyback graph: its edges are the components that differ from j on exactly two cores.

3 Main result

Theorem 2 (Order of a learned saddle)**.** *Let W^* be a pattern teacher, θ_S a learned saddle of component j with $k^* \geq 2$, and assume*

(a) (genericity) $\Lambda_\ell \neq s_j$ for every $\ell < k^*/2$;

(b) (bond dimension) $\chi_e \geq 2$ on the bonds of $\mathrm{St}(V_r)$ for some r with $k_r = k^*$; and, if $\ell^* < \infty$, some node set X of $\mathcal{G}_{\ell^*}$ with spectral radius above s_j satisfies, on every bond e, $\chi_e - 1 \geq \#\{(B, \pi) \in X : |B| \geq 2,\ e \in \mathrm{St}(B)\}$.

Then

$$\mathrm{ord}(\theta_S) = \min(k^*, 2\ell^*). \tag{6}$$

The theorem combines three bounds, proved in Section 5:

- ord $\le k^*$: the *route curve*, new entries on $\mathrm{St}(V_r)$, gives $L - L_S = -s_r s_j^{(K-k^*)/K} t^{k^*} + O(t^{2k^*})$ (Proposition 7);
- ord $\ge \min(k^*, 2\ell)$ whenever $\Lambda_{\ell-1} < s_j$ (Proposition 9); this holds for general residuals, not only pattern teachers;
- ord $\le 2\ell$ whenever $\Lambda_\ell > s_j$ and $\ell < k^*/2$ (Proposition 10).

Assumption (a) excludes the boundary cases $\Lambda_\ell = s_j$, where the bounds do not meet; (b) provides the unused bond channels that the descent curves need.

Corollary 3. *Under the assumptions of Theorem 2:*

(i) If $k^* = 2$, the saddle is strict.

(ii) If $k^* \in \{3, 4\}$, the order is k^* if $\Lambda_1 < s_j$ and 2 if $\Lambda_1 > s_j$. Equivalently, the order is k^* unless the Hessian has a negative eigenvalue; the Hessian block of rotations of single learned vectors is $s_j^{(K-2)/K}(s_j I - \Gamma^{(1)})$.

(iii) If W^* has two components and the larger is learned ($s_r < s_j$), the order is $k^* = k_r$ on every tree.

(iv) If W^* has two components and the smaller is learned ($s_r > s_j$), the order is $2\ell^*$ with $\ell^* = \min \max(\deg B_1, \deg B_2)$ over splits $V_r = B_1 \sqcup B_2$, if this is below $k^*/2$, and k^* otherwise.

(v) In general, the order can be below k^* while the Hessian is positive semidefinite only if $k^* \ge 5$.

Example 4 (Order below k^* with a positive semidefinite Hessian)**.** MPS with six cores, $d = \chi = 2$. Learned $j = 000000$ ($s_j = 1$); remaining $r_1 = 110001$ and $r_2 = 110010$ with weight 0.8 each. Then $V_{r_1} = \{0, 1, 5\}$ ($k = 6$) and $V_{r_2} = \{0, 1, 4\}$ ($k = 5$), so $k^* = 5$. No component differs from j on exactly two cores, so $\mathcal{G}_1$ has no edges and $\Lambda_1 = 0$: the Hessian is positive semidefinite. In $\mathcal{G}_2$ the node $(\{0, 1\}, 11)$ is joined to $(\{5\}, 1)$ by r_1 and to $(\{4\}, 1)$ by r_2, a star with $\Lambda_2 = 0.8\sqrt{2} \approx 1.13 > 1$. Hence $\ell^* = 2 < 5/2$ and the order is 4. The descent puts new entries on cores $0, 1$ toward the pattern 110000, which is not a teacher component, and rotates the learned vectors of cores 4 and 5 toward 1; the two pieces together produce both r_1 and r_2. Minimizing $L - L_S$ numerically over the sphere $\|\theta - \theta_S\| = t$ gives slope 4.000 and constant -0.0350, against the leading-order value $-(2 \cdot 0.8^2 - 1)/8 = -0.0350$. With weights 0.6 instead ($\Lambda_2 = 0.85 < 1$) the order is $5 = k^*$.

Consequence for the dynamics. Near a saddle of order p, the slowest part of the escape is the growth of the small amplitudes that realize the order-p descent.

Proposition 5 (Escape time; derived at leading order)**.** *Start gradient flow at θ_S plus a perturbation whose components along the order-p descent have size ε. At leading order in ε, with the remaining variables relaxed adiabatically, the descent amplitude b obeys $\dot{b} = c\, b^{p-1}$ with $c > 0$, so the escape time is*

$$t_{\mathrm{esc}} \simeq \begin{cases} \ln(1/\varepsilon)/c, & p = 2, \\ \varepsilon^{-(p-2)}/((p-2)c), & p \ge 3. \end{cases} \tag{7}$$

For the route ($p = k^* = k_r$, balanced), $c = s_r s_j^{(K-k_r)/K}$.

In a full run from small initialization the perturbation at the start of the plateau is of order σ, so the plateau after the first escape lasts $\propto \sigma^{-(p-2)}$ (or $\propto \ln(1/\sigma)$). For two components learned in

weight order ($p = k_r$), integrating the route's growth during the first plateau gives

$$\Delta t_2 \simeq \frac{1 - s_r/s_j}{s_r s_j^{(K-2)/K}} \ln \frac{1}{\sigma} \quad (k_r = 2), \qquad \Delta t_2 \simeq \frac{1 - s_r/s_j}{(k_r - 2)\, s_r s_j^{(K-k_r)/K}} \, \sigma^{-(k_r-2)} \quad (k_r \geq 3), \tag{8}$$

where the second formula assumes an initialization aligned with the teacher; the factor $1 - s_r/s_j$ is the growth of the route during the first plateau.

Evidence.

- *Order.* Minimizing $L - L_S$ over spheres of radius t in 32 instances (MPS, binary and HT trees, $k^* = 2$ to 7, pattern and random residuals) gives slopes within 0.003 of the predicted order in the 30 instances the theorem covers; 12 further instances computed with independent code agree within $5 \cdot 10^{-4}$.
- *Plateaus with $p = k^*$.* About 11,600 gradient-flow runs of two-component teachers on nine trees ($K = 3$ to 15) give per-seed exponents of Δt_2 equal to $k_r - 2$ for $k_r = 3, 4, 5$, and for $k_r = 6, 7$ on two trees with balanced initialization. For $k_r = 2$ the logarithmic law fits better than a power law in all 14 settings, with slope 0.98–1.01 against the predicted 1 (8). Off $\mathrm{St}(V_r)$, the second singular value of every bond stays below $4.3\,\sigma^2 s_r$ in all 5,131 learning events.
- *Plateaus with $p = 2\ell^* < k^*$.* A pre-registered test (about 2,800 runs) started at θ_S plus a perturbation of size ε measured $t_{\mathrm{esc}} \propto \varepsilon^{-\gamma}$ with $\gamma = 2.001 \pm 0.001$ for $p = 4$ (Example 4, and an MPS6 two-component case with $k^* = 6$) and $\gamma = 3.996$–3.998 for $p = 6$ (HT and binary trees, $k^* = 7$), wherever ε lay below the size at which the order-p descent is predicted to beat the route; the route law $\gamma = k^* - 2$ fits 10^3–10^5 times worse. Above that size the results were mixed. This test has not been independently re-checked.

4 Relation to prior work

Deep linear networks. Saddle-to-saddle learning from small initialization, the exponent $K - 2$ of the first escape, and greedy order are established for DLNs (Saxe et al., 2014; Jacot et al., 2022); the later saddles are conjectured there. Achour et al. (2024, Thm 7) classify DLN critical points as strict or non-strict, which is a lower bound of the form "order ≥ 3". The binary tree with four sites is exactly a depth-3 DLN, so Theorem 2 reduces there to known facts: $k^* = 3$, and the condition $\Lambda_1 \lesseqgtr s_j$ is the DLN criterion for strict saddles.

Homogeneous models. Near the origin, the weights of a homogeneous model align with a KKT point of the correlation with the target and escape after a time $\propto \sigma^{-(K-2)}$ (Kumar and Haupt, 2025). Escapes that need one, two, or $k \geq 3$ small layers take $O(1)$, logarithmic and $\propto \sigma^{-(k-2)}$ time (Rawal and DeWeese, 2026, Thm 4.2), and alternating gradient flows carry dormant amplitudes across plateaus (Kunin et al., 2025), which gives head starts as in (8) (Zhang et al., 2026, Thm 4 for two-layer networks). These results take the number of small factors as given. Theorem 2 determines it for TTNs.

The landscape of TTNs. Furman et al. (2026) study the landscape of exactly this model. They prove that minimum-norm local minima are global, and that at a minimum-norm critical point every connected set of rank-deficient bonds (model rank below teacher rank) with N incident

cores yields a straight-line descent of order N (their Lemmas D.8–D.9), which bounds the order from above. They remark that with several disjoint such sets the smallest gives the fastest escape and relate N to the leap of Abbe et al. (2023); they prove one lower bound, for a parity saddle (their Prop. 5.3), and do not treat dynamics.

What is new here.

(1) *The exact order.* A matching lower bound at every learned saddle with one learned component, so the order is determined, not only bounded. Previously lower bounds were known for DLNs and one parity saddle.

(2) *The relevant set is the smallest Steiner subtree.* For a two-component teacher the rank-deficient bonds are exactly the bonds of $\mathrm{St}(V_r)$, so the upper bound of Furman et al. (2026) already equals k^* there and our contribution is the lower bound. With several remaining components the rank-deficient bonds form one connected set, the union of all their Steiner subtrees, often the whole tree; the order is set by the smallest single Steiner subtree, which can be much smaller.

(3) *Orders below k^*.* The spectral condition, the order $2\ell^*$, and the descent through non-teacher patterns are new; they show that "smallest set of cores that must move" is not the whole story from $k^* = 5$ on.

(4) *Dynamics.* The escape laws (7)–(8) with their TTN constants, and their tests.

Frandsen and Ge (2022) exhibit descents of order 2, 3 and 4 at balanced points of Tucker decompositions, without lower bounds.

5 Proofs

Throughout, θ_S is a learned saddle of component j, $\varepsilon = \|\theta - \theta_S\|$, and ν_v denotes the vector of new entries of core v.

5.1 Sectors

Let $P_v = q_v q_v^\top$, $P_v^\perp = I - P_v$, and for $B \subseteq V$

$$\Pi_B = \bigotimes_{v \in B} P_v^\perp \otimes \bigotimes_{v \notin B} P_v.$$

The 2^K projectors Π_B are orthogonal and sum to the identity. Write $\mathcal{H}_B$ for the range of Π_B; then $\mathcal{H}_{B_1} \otimes \mathcal{H}_{B_2} \cong \mathcal{H}_{B_1 \sqcup B_2}$ for disjoint B_1, B_2, and a basis of $\mathcal{H}_B$ is given by the patterns on the sites of B that differ from j on every core of B. Let $\mathcal{R}_B = \Pi_B \mathcal{R}$. For a pattern teacher,

$$\mathcal{R}_B = \sum_{r:\, V_r = B} s_r U_r, \tag{9}$$

since U_r lies in the sector of the cores on which it differs from j. Hence $k^* = \min\{\deg B : \mathcal{R}_B \neq 0\}$.

Lemma 6 (Criticality and the energy identity)**.** *(i) θ_S is critical iff $\mathcal{R}_{\{v\}} = 0$ for every v, i.e. iff $k^* \geq 2$. (ii) Let $N(\theta) = \langle U_j, W(\theta) \rangle$ and, for $B \neq \emptyset$, $m_B = \Pi_B W/N$. Then, wherever $N \neq 0$,*

$$L(\theta) - L_S = \tfrac{1}{2}(N - s_j)^2 + \sum_{B \neq \emptyset} \Big[\tfrac{N^2}{2} \|m_B\|^2 - N \langle \mathcal{R}_B, m_B \rangle \Big]. \tag{10}$$

Proof. (i) A new entry meets old entries with index 0 on its bond in the neighbouring core, so it does not change W at first order; an old entry of core v moves W within $\mathcal{H}_\emptyset \oplus \mathcal{H}_{\{v\}}$. The gradient therefore vanishes iff $\mathcal{R}$ has no component in any $\mathcal{H}_{\{v\}}$ (its component in $\mathcal{H}_\emptyset$ is zero by definition of s_j). (ii) By orthogonality $\|W - W^*\|^2 = \sum_B \|\Pi_B W - \Pi_B W^*\|^2$, with $\Pi_\emptyset(W - W^*) = (N - s_j)U_j$, $\Pi_B W^* = \mathcal{R}_B$ and $\Pi_B W = N m_B$ for $B \neq \emptyset$, and $L_S = \frac{1}{2}\sum_B \|\mathcal{R}_B\|^2$. □

5.2 Upper bound k^*: the route curve

Proposition 7. *Let r have $k_r = k^*$ and $\chi_e \geq 2$ on the bonds of $\mathrm{St}(V_r)$. Give each core $v \in \mathrm{St}(V_r)$ the new entry $t \cdot$ (pattern of r on the sites of v) with bond index 1 on the bonds inside $\mathrm{St}(V_r)$ and 0 on the other bonds. At the balanced saddle ($a_v = s_j^{1/K}$),*

$$L(\theta(t)) - L_S = -s_r\, s_j^{(K-k^*)/K}\, t^{k^*} + \tfrac{1}{2} s_j^{2(K-k^*)/K}\, t^{2k^*},$$

and $\|\theta(t) - \theta_S\| = \sqrt{k^*}\, t$. Hence $\mathrm{ord}(\theta_S) \leq k^*$.

Proof. Expand the contraction core by core into old and new entries. A term that uses a new entry on some core of $\mathrm{St}(V_r)$ but an old entry on a neighbouring core of $\mathrm{St}(V_r)$ contracts index 1 with index 0 and vanishes. So only two terms survive: all old entries, giving $s_j U_j$, and new entries on all of $\mathrm{St}(V_r)$ with old entries elsewhere, giving $t^{k^*} s_j^{(K-k^*)/K} U$, where U has the pattern of r on the sites of $\mathrm{St}(V_r)$ and that of j elsewhere. Since r agrees with j outside V_r, $U = U_r$. Thus $W - W^* = -\mathcal{R} + t^{k^*} s_j^{(K-k^*)/K} U_r$, and $\langle \mathcal{R}, U_r \rangle = s_r$, $\|U_r\| = 1$. □

5.3 Lower bound: locality of cumulants

Cumulants. Consider families $f = (f_B)_{B \subseteq V}$ with $f_B \in \mathcal{H}_B$, with the product $(f * g)_B = \sum_{B_1 \sqcup B_2 = B} f_{B_1} \otimes g_{B_2}$. This is a commutative associative algebra with unit $(1, 0, 0, \dots)$, and every f with $f_\emptyset = 0$ is nilpotent, so exp and log are finite series and $\log(f * g) = \log f + \log g$. With $m_\emptyset = 1$ define the cumulants $\kappa = \log m$. Explicitly, by the set-partition formulas,

$$m_B = \sum_{\pi \in \mathrm{Part}(B)} \bigotimes_{C \in \pi} \kappa_C, \qquad \kappa_B = \sum_{\pi \in \mathrm{Part}(B)} (-1)^{|\pi|-1} (|\pi| - 1)! \bigotimes_{C \in \pi} m_C.$$

All m_B and κ_B are real-analytic near θ_S and vanish at θ_S for $B \neq \emptyset$, so $\|\kappa_B\| \leq C_0 \varepsilon$.

Lemma 8 (Locality)**.** *For $|B| \geq 2$, near θ_S,*

$$\|\kappa_B(\theta)\| \leq C_0 \prod_{x \in \mathrm{St}(B)} \|\nu_x\|.$$

In particular $\|\kappa_B\| \leq C_0 \varepsilon^{\deg B}$.

Proof. Vanishing. Fix $x \in \mathrm{St}(B)$ and suppose $\nu_x = 0$. Then every nonzero entry of G_x has all bond indices 0, so the contraction factorizes at x: $W = g_x \otimes \bigotimes_b W_b$, where g_x is the physical vector of G_x, b runs over the branches of T at x (the components of $T \setminus x$, with node sets V_b), and W_b is the contraction of branch b with its bond to x fixed to 0. The projectors factorize core by core, so $m = m^{(x)} * \prod_b m^{(b)}$, where $m^{(x)}$ is supported on $\{\emptyset, \{x\}\}$ and $m^{(b)}$ on subsets of V_b. Taking logarithms, $\kappa = \log m^{(x)} + \sum_b \log m^{(b)}$ is supported on subsets of $\{x\}$ or of a single V_b. B is none of these: if $x \in B$, then $B \not\subseteq \{x\}$ and $B \not\subseteq V_b$; if $x \notin B$ and $B \subseteq V_b$, the connected set V_b would contain $\mathrm{St}(B) \ni x$. So $\kappa_B = 0$ whenever $\nu_x = 0$ for some $x \in \mathrm{St}(B)$.

Bound. Expand κ_B as a power series in $\theta - \theta_S$. Setting $\nu_x = 0$ removes exactly the monomials that contain a variable of ν_x, and the result is zero; so every monomial contains a variable of ν_x for every $x \in \mathrm{St}(B)$. Factoring one such variable per x out of each monomial and bounding the rest on a small polydisc gives the claim. □

Lemma 8 is the precise sense in which a change of W in sector B "needs new entries on every core of $\mathrm{St}(B)$".

Proposition 9 (Lower bound)**.** *Let θ_S be critical, $2 \le \ell \le \lceil k^*/2 \rceil$, and let $\mathcal{S}_{\ell-1}$ be the operator on $\bigoplus_{\deg C \le \ell-1} \mathcal{H}_C$ with blocks $(x, y) \mapsto \langle \mathcal{R}_{C_1 \sqcup C_2}, x \otimes y \rangle$ for disjoint C_1, C_2 and zero otherwise. If $\lambda_{\max}(\mathcal{S}_{\ell-1}) < s_j$, then $L(\theta) - L_S \ge -C\varepsilon^{\min(k^*, 2\ell)}$ near θ_S. For a pattern teacher $\mathcal{S}_{\ell-1} = \Gamma^{(\ell-1)}$ in the pattern basis, so the condition is $\Lambda_{\ell-1} < s_j$.*

Proof. Let $p = \min(k^*, 2\ell)$, $y_C = \|\kappa_C\|$, and call C *low* if $\deg C \le \ell - 1$ and *high* otherwise. Four facts: (F1) a high set has $\deg C \ge \ell \ge 2$, so $y_C \le C_0 \varepsilon^\ell$ by Lemma 8; (F2) subsets of low sets are low; (F3) low sets carry no residual, since $\deg \le \ell - 1 < k^*$; (F4) summing $\langle \mathcal{R}_B, \kappa_{C_1} \otimes \kappa_{C_2} \rangle$ over all B and all unordered splits $B = C_1 \sqcup C_2$ into low sets gives exactly $\frac{1}{2}\langle \kappa_{\mathrm{low}}, \mathcal{S}_{\ell-1}\kappa_{\mathrm{low}} \rangle \le \frac{1}{2}\lambda_{\max}(\mathcal{S}_{\ell-1})\|\kappa_{\mathrm{low}}\|^2$, where $\|\kappa_{\mathrm{low}}\|^2 = \sum_{C \text{ low}} y_C^2$.

Quadratic term. For low B, $m_B - \kappa_B$ is a sum of products of at least two cumulants of low sets (F2), each bounded by $C_0 \varepsilon\, y_C$ for one of its factors. Hence $\sum_{B \text{ low}} \|m_B\|^2 \ge (1 - C_2\varepsilon)\|\kappa_{\mathrm{low}}\|^2$. The high-$B$ terms of the quadratic sum are nonnegative and are dropped.

Linear term. Only B with $\mathcal{R}_B \ne 0$ contribute, and then $\deg B \ge k^*$. Expand m_B over partitions: the single cumulant gives $|\langle \mathcal{R}_B, \kappa_B \rangle| \le C\varepsilon^{k^*}$; two low blocks give (F4); one low and one high block give at most $\frac{\gamma}{2} y_{C_1}^2 + C_\gamma \varepsilon^{2\ell}$ (Young's inequality and F1); two high blocks give $O(\varepsilon^{2\ell})$; partitions with three or more blocks give either $O(\varepsilon)(y_{C_1}^2 + y_{C_2}^2)$ (two low blocks) or $O(\varepsilon^{2\ell})$ (two high blocks). Summing,

$$\sum_B \langle \mathcal{R}_B, m_B \rangle \le \tfrac{1}{2}\big(\lambda_{\max}(\mathcal{S}_{\ell-1}) + M\gamma + C\varepsilon\big)\|\kappa_{\mathrm{low}}\|^2 + C_\gamma \varepsilon^p.$$

Conclusion. By (10), dropping $\frac{1}{2}(N - s_j)^2 \ge 0$,

$$L - L_S \ge \tfrac{N}{2}\big[N(1 - C_2\varepsilon) - \lambda_{\max}(\mathcal{S}_{\ell-1}) - M\gamma - C\varepsilon\big]\|\kappa_{\mathrm{low}}\|^2 - NC_\gamma\varepsilon^p.$$

With $g = s_j - \lambda_{\max}(\mathcal{S}_{\ell-1}) > 0$, $\gamma = g/(4M)$ and $N \ge s_j - C\varepsilon$, the bracket is positive for small ε, so $L - L_S \ge -C\varepsilon^p$. □

The proof uses only the sector decomposition, criticality and multilinearity. It does not use balance, the bond dimensions, or the pattern structure of $\mathcal{R}$.

5.4 Upper bound 2ℓ: polymer curves

Proposition 10 (Sharpness)**.** *Let W^* be a pattern teacher, $\ell < k^*/2$, and X a node set of $\mathcal{G}_\ell$ whose adjacency matrix A_X has spectral radius $\rho > s_j$, with the bond-dimension condition of Theorem 2(b). Then there is a curve with $\|\theta(t) - \theta_S\| \le t$ and $L(\theta(t)) - L_S \le -ct^{2\ell}$.*

Proof. Work at the balanced saddle, $a_v = a = s_j^{1/K}$. Let $x \ge 0$ be a unit Perron vector, $A_X x = \rho x$. For $0 < \eta \le 1$:

- for a node $(\{v\}, \pi)$, set the old entry $G_v[\pi, 0, \dots, 0] = a\,\eta\, x_{(\{v\},\pi)}$ (a rotation of the learned vector of v);

- for a node $n = (B, \pi)$ with $|B| \geq 2$, let $P = \mathrm{St}(B)$ and $h = |P| \leq \ell$. Choose on every bond inside P a new index value used by no other multi-core node of X on that bond (possible by the bond-dimension condition), and give each core $u \in P$ the new entry $a(\eta x_n)^{1/h}$ with physical index π on $u \in B$ and that of j on $u \in P \setminus B$, the chosen indices on the bonds of P, and 0 on its other bonds.

Then $\|\theta - \theta_S\| \leq C\eta^{1/\ell}$. In a nonzero term of the contraction, each connected set of bonds carrying nonzero indices uses the entries of a single multi-core node n and equals its $\mathrm{St}(B)$. So W is a sum over collections of nodes ("polymers" and rotations) using disjoint cores, each contributing the factor ηx_n times its pattern, with a on every other core. In particular $N = s_j$ exactly, and $m_B = \eta m_B^{(1)} + \eta^2 m_B^{(2)} + O(\eta^3)$, where $m^{(1)}$ collects single nodes and $m^{(2)}$ pairs of nodes on disjoint cores.

Order η. $m_B^{(1)} = \sum_\pi x_{(B,\pi)} e_\pi$, so $\sum_B \|m_B^{(1)}\|^2 = \|x\|^2 = 1$. No single node has $B = V_r$ (that would need $\deg B \geq k^* > \ell$), so $\sum_B \langle \mathcal{R}_B, m_B^{(1)} \rangle = 0$ by (9).

Order η^2. For an edge between (B_1, π_1) and (B_2, π_2) from component r, the sets $\mathrm{St}(B_1)$ and $\mathrm{St}(B_2)$ are disjoint; otherwise their union would be a connected set containing V_r with at most $2\ell - 1 < k^*$ cores. So every edge is a pair on disjoint cores, and $\sum_B \langle \mathcal{R}_B, m_B^{(2)} \rangle = \sum_{\text{edges}} s_r x_{n_1} x_{n_2} = \frac{1}{2} x^\top A_X x = \frac{\rho}{2}$.

By (10) with $N = s_j$, $L - L_S = \frac{s_j^2}{2}\eta^2 - s_j \frac{\rho}{2}\eta^2 + O(\eta^3) = -\frac{s_j}{2}(\rho - s_j)\eta^2 + O(\eta^3)$. Setting $\eta = (t/C)^\ell$ gives the claim. □

5.5 Proof of Theorem 2 and Corollary 3

Case $\ell^* = \infty$. Then $\Lambda_\ell < s_j$ for all $\ell < k^*/2$ by (a), in particular for $\ell = \lceil k^*/2 \rceil - 1$. Proposition 9 with $\ell = \lceil k^*/2 \rceil$ gives ord $\geq \min(k^*, 2\lceil k^*/2 \rceil) = k^*$, and Proposition 7 gives ord $\leq k^*$. (If $k^* = 2$ the order is 2 directly: the route curve has order 2 and every critical point has order at least 2.)

Case $\ell^* < \infty$. If $\ell^* = 1$, Proposition 10 gives order ≤ 2, hence $= 2$. If $\ell^* \geq 2$, then $\Lambda_{\ell^*-1} < s_j$ by minimality and (a), so Proposition 9 at level ℓ^* gives ord $\geq \min(k^*, 2\ell^*) = 2\ell^*$, and Proposition 10 gives ord $\leq 2\ell^*$. In both cases ord $= \min(k^*, 2\ell^*)$. □

Corollary 3. (i) $\ell^* < k^*/2 = 1$ is impossible. (ii) For $k^* \in \{3, 4\}$ the only level below $k^*/2$ is $\ell = 1$. Rotating the learned vector of core u by ε_u toward π changes W by a mixed pattern of norm $a^{K-1}\varepsilon_u$; two rotations couple through a component that differs from j on exactly those two cores, with coefficient $-s_r a^{K-2}$. So the rotation block is $a^{2(K-1)} I - a^{K-2}\Gamma^{(1)} = s_j^{(K-2)/K}(s_j I - \Gamma^{(1)})$. (iii) With one remaining component every node of $\mathcal{G}_\ell$ has at most one edge, so $\Lambda_\ell \in \{0, s_r\}$, which is below s_j. (iv) Same observation with $s_r > s_j$: $\Lambda_\ell = s_r > s_j$ from the first level that admits a split. (v) For $k^* \leq 4$, by (i)–(ii) an order below k^* requires $\Lambda_1 > s_j$, which makes the Hessian indefinite. □

5.6 Escape time (Proposition 5)

Route. With old entries frozen at θ_S and route amplitudes b_v on $\mathrm{St}(V_r)$, the computation of Proposition 7 gives $\dot{b}_v = A \prod_{u \neq v} b_u \, (s_r - A \prod_u b_u)$ with $A = a^{K-k}$. For a balanced route with $b \ll 1$, $\dot{b} = s_r a^{K-k} b^{k-1}$, which is $\dot{b} = c b^{p-1}$ with $p = k$; integrating $\frac{d}{dt} b^{-(p-2)} = -(p-2)c$ gives (7).

Polymer. In the setting of Proposition 10 with one multi-core node, the rotations have $O(1)$ curvature and follow the polymer amplitude $y = \prod_{v \in P} b_v$ adiabatically. Eliminating them leaves $L_{\text{eff}} = -\frac{c}{2} y^2$ with $c > 0$ (for Example 4, $c = s_{r_1}^2 + s_{r_2}^2 - s_j^2$ in units $s_j = 1$), so $\dot{b}_v = c \, b_v \prod_{u \neq v} b_u^2$, and for balanced amplitudes $\dot{b} = c b^{2\ell - 1} = c b^{p-1}$.

Head start. During the first plateau the learned cores grow from σ as $\dot{a} \simeq s_j a^{K-1}$, and the route already grows with coupling $s_r a^{K-k}$. For $k \geq 3$, $\frac{d}{dt} b^{-(k-2)} = -(k-2) s_r a^{K-k}$ and $\int a^{K-k} dt = \int a^{1-k} da / s_j = \sigma^{-(k-2)} / ((k-2) s_j) + O(1)$, so the route enters the plateau with $b^{-(k-2)} = \sigma^{-(k-2)}(1 - s_r/s_j) + O(1)$. For $k = 2$, $\ln b$ grows by $(s_r/s_j) \ln(1/\sigma) + O(1)$. Inserting into (7) gives (8).

6 Scope

The theorem covers saddles with one learned component, pattern teachers for the exact equality (the lower bound holds for any residual), whitened inputs and the population loss. Saddles with several learned components and the boundary cases $\Lambda_\ell = s_j$ are open. Proposition 5 is a leading-order derivation, not a theorem; when $2\ell^* < k^*$ both mechanisms are present, and the lower order wins only below a crossover scale set by the prefactors.

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